

COMP 3311: Database Management Systems

Lab 9 Exercise: Oracle PL/SQL and Stored Procedures

WHAT TO DO

1. **Download** the zipped folder **Lab9Exercise.zip** from the *Lab 9* entry of the *Lab Assignments* section of the *Assignments* tab on Canvas to the desktop and **unzip** it. The folder contains two SQL script files **lab9db.sql** and **lab9Queries.sql** and a text file **lab9CgaCalculations.txt**.
 2. **Place** your **insertMyself.sql** script file **inside** the **Lab9Exercise** folder and **modify** it so that it inserts the value **null** for the **cga** attribute. **All other values should be the same as those for Lab 8.** ← **IMPORTANT!**
 3. **Execute** the **lab9db.sql** script file in **Oracle SQL Developer**.
 4. In the **Connections** pane of **SQL Developer**, expand your connection, right-click the **Procedures** node and select **New Procedure...** from the context menu. In the **Create Procedure** dialog name the procedure **lab9CgaCalculations** and select the **OK** button.
 5. **Replace** all the code in your **lab9CgaCalculations** procedure with the code in the file **lab9CgaCalculations.txt**.
 6. **Complete** the **TODOs** in the **lab9CgaCalculations** procedure to do the following.
 - a. Complete the declaration of the **all** the variables needed for the **cga** calculation as well as replace with the correct **select** statement the declaration of the **enrollsInCursor** for the **EnrollsIn** table.
 - b. For each student record retrieved by the **studentCursor**
 - i. retrieve **only** the current student's related **EnrollsIn** records (**do not** retrieve all **EnrollsIn** records) and collect the data needed to calculate the current student's **cga**.
 - ii. calculate the student's **cga** to two decimal places according to the formula given below and update the student's record with the calculated **cga**.
 - iii. insert the record into the **LowCga** table if the **cga** is less than or equal to 2. The **LowCga** table is created by the **lab9db.sql** script file.
- Note:** The command **dbms_output.put_line(<string>)** outputs **<string>** to the Script Output tab where **<string>** is replaced with any character string enclosed in single quotes (see the script file for an example).
7. **After your lab9CgaCalculations procedure executes correctly, execute** the **lab9Queries.sql** script file in **Oracle SQL Developer**. The **lab9Queries.sql** script file contains the following code.

```
clear screen
set serveroutput on
set pagesize 30
set termout off
@lab9db
set termout on
set feedback off
```

The **set** commands do the following:

- **set serveroutput on** - enables display of the **dbms_output.put_line** command argument;
- **set pagesize 30** - sets the output page size to 30 lines;
- **set termout off** - disables display to the Script Output tab;
- **set termout on** - enables display to the Script Output tab.

```
-- execute the lab9CgaCalculations procedure
exec lab9CgaCalculations;
select studentId, firstName, lastName, cga from Student order by cga desc nulls last;
select studentId, firstName, lastName, cga from LowCga order by cga desc nulls last;
```

The final two lines display the result of calculating each student's **cga** as well as those students whose **cga** is less than or equal to 2.

CGA CALCULATION

A student's **cga** is calculated according to the following formula:

$$cga = \frac{\sum(\text{course credit} \times \text{course grade point})}{\sum \text{course credit}}$$

For each course in which the student is enrolled, multiply the course credit by the course grade point and sum these values for all the courses. Then, divide the previous sum by the sum of the course credit of all the courses in which the student is enrolled.

Lab 9 Exercise: Oracle PL/ SQL and Stored Procedures

To obtain the grade point for a course, convert the course grade to a grade point according to the following formula:

$$\text{course grade point} = \text{maximum}((\text{grade} / 20) - 1, 0)$$

Note that the code to do the grade to grade point conversion is already given in the file **lab9CgaCalculations**.

WHAT TO SUBMIT

1. Your completed **lab9CgaCalculations** procedure code as a text file. You can copy and paste your code from **Oracle SQL Developer** into a text (txt) file.
2. A screenshot of the **Oracle SQL Developer** window or a text file that shows the result of running the **lab9Queries.sql** script file in the Script Output tab as shown in Figure 1.

```
>>> Steve Jobs (15000655) with cga=3.56 is an honours Student.  
>>> Alan Turing (28918856) with cga=3.56 is an honours Student.  
>>> Typical Student (11111111) with cga=3.56 is an honours Student.
```

Your record should be shown in the top message and in the result of the first query.

STUDENTI	FIRSTNAME	LASTNAME	UNIT	CGA
11111111	Typical	Student	COMP	3.56
15000655	Steve	Jobs	COMP	3.56
28918856	Alan	Turing	MATH	3.56
26186666	Warren	Buffet	MGMT	3.42
15085942	Bill	Gates	COMP	3.4
29873381	Nikola	Tesla	ELEC	3.37
13556789	Legolas	Greenleaf	MATH	3.36
13782973	Edith	Clarke	ELEC	3.15
28834512	Isaac	Newton	MATH	2.98
15678901	Albert	Einstein	COMP	2.97
13456789	Ariana	Grande	COMP	2.82
13455789	Harry	Potter	COMP	2.76
18792018	Elon	Musk	MGMT	2.76
15678989	Edith	Clarke	COMP	2.73
15456789	Leonardo	Da Vinci	COMP	2.72
16789012	Robert	Redford	MATH	2.57
26184624	Bruce	Wayne	ELEC	2.47
66666666	Ferris	Bueller	MGMT	1.54
28435018	David	Decker	MGMT	1.49
26184444	Donald	Trump	MGMT	1.49
14567890	David	Decker	ELEC	1.49
99987654	Lazzy	Lazy	COMP	
27182481	Buzzy	Bizy	MGMT	
STUDENTI	FIRSTNAME	LASTNAME	UNIT	CGA
66666666	Ferris	Bueller	MGMT	1.54
28435018	David	Decker	MGMT	1.49
14567890	David	Decker	ELEC	1.49
26184444	Donald	Trump	MGMT	1.49

Figure 1: Example **SQL Developer** Script Output tab showing the result of executing the **lab9Queries.sql** script file.

This is an individual exercise. Each student must submit their own exercise files.

HOW TO SUBMIT

By 6 p.m. tomorrow, upload your completed **lab9CgaCalculations.txt** text file and the screenshot or text file showing the result of running the **lab9Queries.sql** script file to Canvas.

Note: You will only be able to submit files with extension txt, jpg or jpeg.